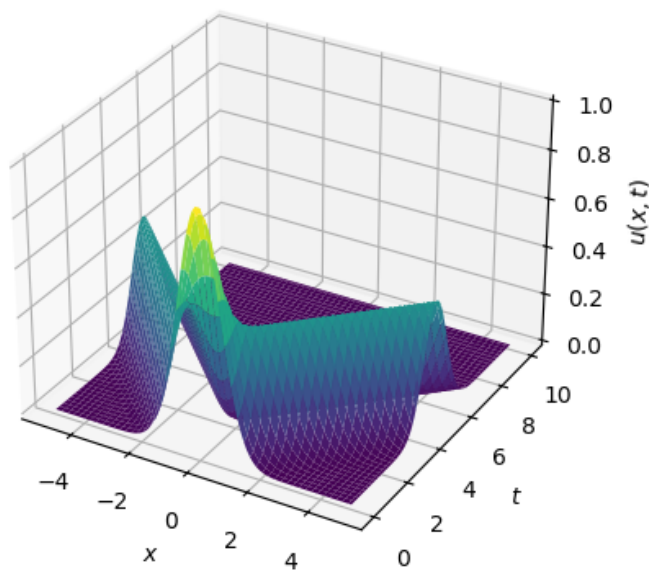


Question 1

(a) Given $\phi(x) = e^{-x^2}$, $\psi(x) = 0$

$$\begin{aligned} u(x, t) &= \frac{1}{2} \left(e^{-(x-ct)^2} + e^{-(x+ct)^2} \right) + \frac{1}{2c} \int_{x-ct}^{x+ct} 0 ds \\ &= \frac{1}{2} \left(e^{-(x-ct)^2} + e^{-(x+ct)^2} \right) \end{aligned}$$

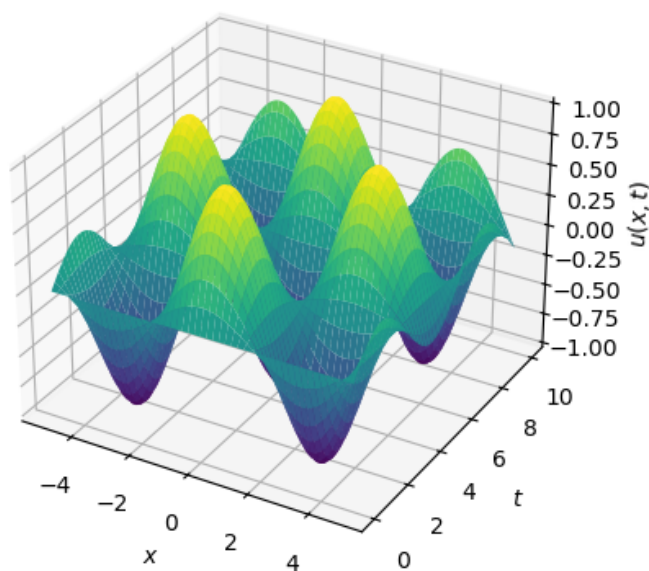
D'Alembert soln for $\phi(x) = e^{-x^2}$ and $\psi(x) = 0$ with $c = 1$



(b) Given $\phi(x) = 0$, $\psi(x) = \cos x$

$$\begin{aligned} u(x, t) &= \frac{1}{2} (0 + 0) + \frac{1}{2c} \int_{x-ct}^{x+ct} \cos(s) ds \\ &= \frac{1}{2c} \left[\sin(s) \right]_{x-ct}^{x+ct} \\ &= \frac{1}{2c} \left[\sin(x + ct) - \sin(x - ct) \right] \end{aligned}$$

D'Alembert soln for $\phi(x) = 0$ and $\psi(x) = \cos x$ with $c = 1$



(c) Given $\phi(x) = \begin{cases} 1 & |x| < 1 \\ 0 & \text{otherwise} \end{cases}$, $\psi(x) = 0$

For $|x| < 1$,

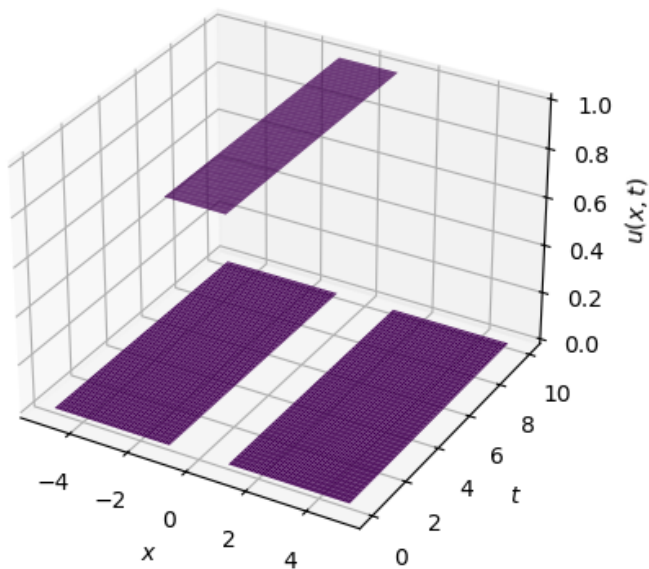
$$\begin{aligned} u(x, t) &= \frac{1}{2}(1 + 1) \\ &= 1 \end{aligned}$$

For $|x| > 1$

$$u(x, t) = 0$$

$$u(x, t) = \begin{cases} 1 & |x| < 1 \\ 0 & \text{otherwise} \end{cases}$$

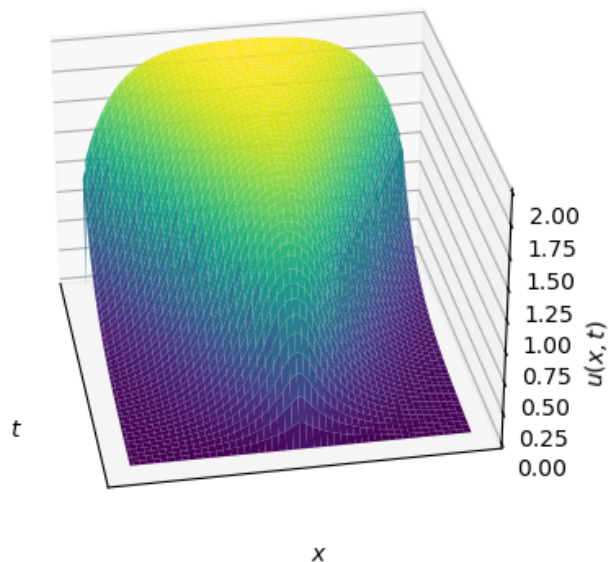
D'Alembert soln for $\phi(x) = 1$ if $|x| < 1$ and 0 otherwise, and $\psi(x) = 0$ with $c = 1$



(d) Given $\phi(x) = 0$, $\psi(x) = e^{-|x|}$

$$\begin{aligned}
 u(x, t) &= \frac{1}{2c} \int_{x-ct}^{x+ct} e^{-|s|} ds \\
 &= \begin{cases} \frac{1}{2c} \int_{x-ct}^{x+ct} e^s ds & s < 0 \\ \frac{1}{2c} \int_{x-ct}^{x+ct} e^{-s} ds & s > 0 \end{cases} \\
 &= \begin{cases} \frac{1}{2c} \left[e^s \right]_{x-ct}^{x+ct} & s < 0 \\ \frac{1}{2c} \left[-e^{-s} \right]_{x-ct}^{x+ct} & s > 0 \end{cases} \\
 &= \begin{cases} \frac{1}{2c} \left[e^{x+ct} - e^{x-ct} \right] & x < -ct \\ \frac{1}{2c} \left[-e^{-(x+ct)} - e^{-(x-ct)} \right] + \frac{1}{c} & -ct < x < ct \\ \frac{1}{2c} \left[-e^{-(x+ct)} + e^{-(x-ct)} \right] & x > ct \end{cases}
 \end{aligned}$$

D'Alembert soln for $\phi(x) = 0$ and $\psi(x) = e^{-|x|}$ with $c = 1$



Question 2

(a)

Question 3

Question 4

Question 5